Do Now · retrieval

Fetch–decode–execute and buses

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which register holds the address accessed during a fetch?

Q2

What does the CIR hold?

Q3

What kind of information travels along the address bus?

Q4

A rising-edge D flip-flop has Q=0. D changes to 1 between clock edges; the next rising edge occurs while D=1. State Q before and after that edge, and explain why.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Fetch uses the buses — The address bus carries the address selected by MAR; memory returns instruction data on the data bus; control signals coordinate the transfer.

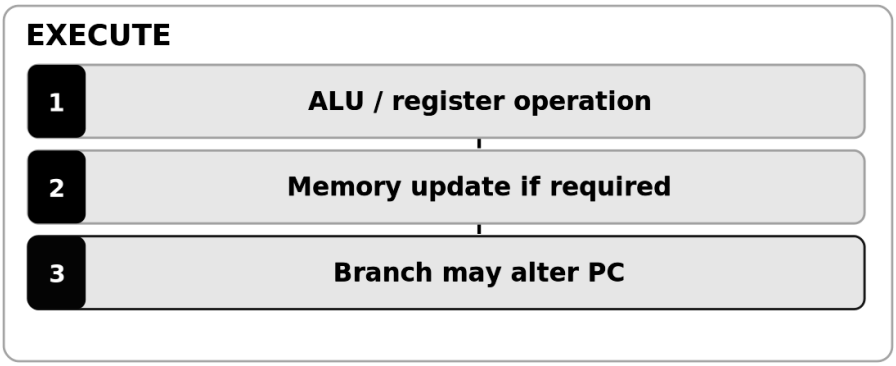
Core idea 2

Decode interprets instruction — The control unit examines CIR to decide which operation and operands are involved.

Core idea 3

Execute changes state — An instruction may update ACC or memory, or alter PC for a jump. Interrupt handling is coordinated at appropriate points and can temporarily redirect execution.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

fetch — Retrieve next instruction from memory.

decode — Interpret opcode and operands.

execute — Perform the operation.

address bus — Transfers memory/device addresses.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

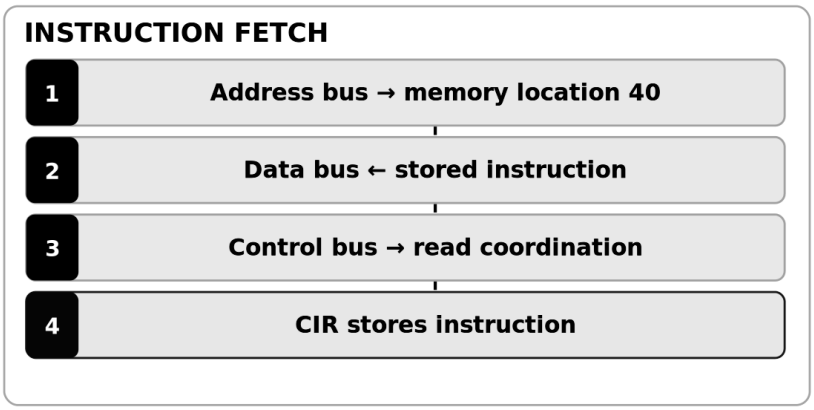
Fetch the instruction at address 40 and explain what each bus contributes.

Follow the worked method

1. Predict first — MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

2. Explain the answer — The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

The initial fetch advances PC from 81 to 82, as in a sequential instruction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For each bus—address, data and control—state its main function and give one example of the information transported.

Question 2

Trace a fetch starting PC=64, then explain what happens when CIR contains JMP 100.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare what happens in the instruction cycle during a normal arithmetic instruction and while responding to an interrupt.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An unconditional branch can change the Program Counter away from its sequential value.

Use key vocabulary from memory

1. Explain fetch in your own words.

2. Explain decode in your own words.

3. Explain execute in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.